COMP 597 Project

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Workload Overview: PNA on PCQM4Mv2

PNA (Principal Neighbourhood Aggregation) is a graph neural network (GNN) architecture designed for learning on graph data.

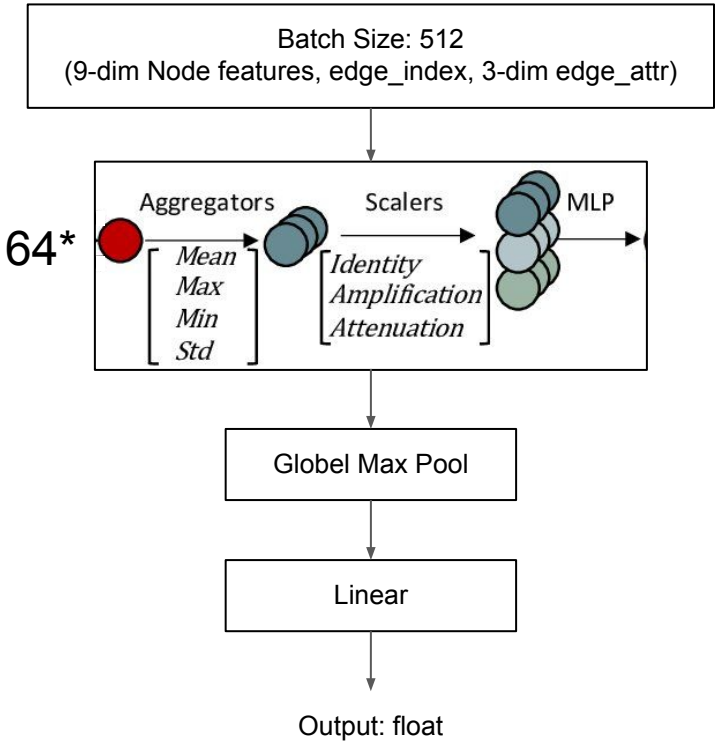
What is it used for?

PNA is used for **graph-level prediction tasks**, such as molecular property prediction. In our workload, it is trained on molecular graphs from the **PCQM4Mv2 dataset**, where:

- Nodes represent atoms,
- Edges represent chemical bonds
- The target is a real-valued molecular property

What is the model structure?

- **64 stacked PNA Conv layers**, which iteratively update node embeddings by aggregating neighbor information.
- **A global max pooling layer**, which converts variable-sized node embeddings into a fixed-size graph embedding
- **A linear layer**, which maps the graph embedding to a single real-valued prediction.



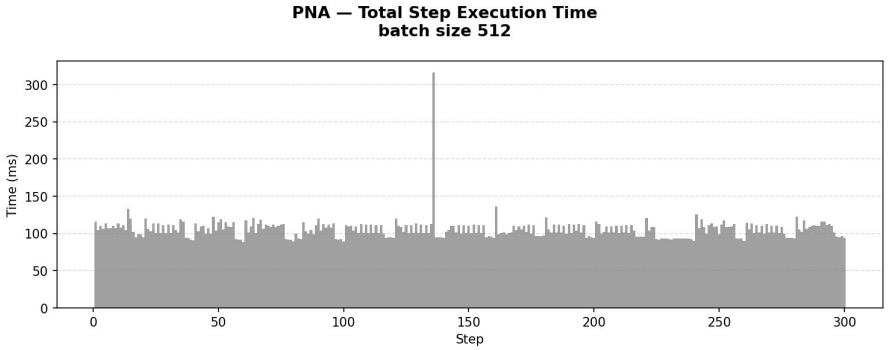
Experiment Setup: Measurements

#	Measurement	Purpose	What It Records
1	**Raw measurement**	Capture the workload as-is, comparing behavior of different batch sizes.	CUDA-synced step timing.
2	**GC-controlled e2e baseline**	Establish a clean baseline free of GC noise. Automatic GC disabled during training; full gen-2 sweep forced between epochs. Foundation for measurements 3 and 4.	Same step & substep timing as raw, but without GC-induced variability.
3	**Hardware utilization**	Quantify GPU, CPU, and memory usage across batch sizes. Built on (2).	GPU util (`pynvml`), per-process CPU util (`psutil`), RAM usage. Sampled at 500 ms intervals.
4	**Energy and carbon**	Measure per-step energy consumption and CO_2 emissions. Built on (2).	CPU/GPU/RAM energy breakdown (kWh) and CO_2 emissions via CodeCarbon `OfflineEmissionsTracker` with 500 ms measurement windows.

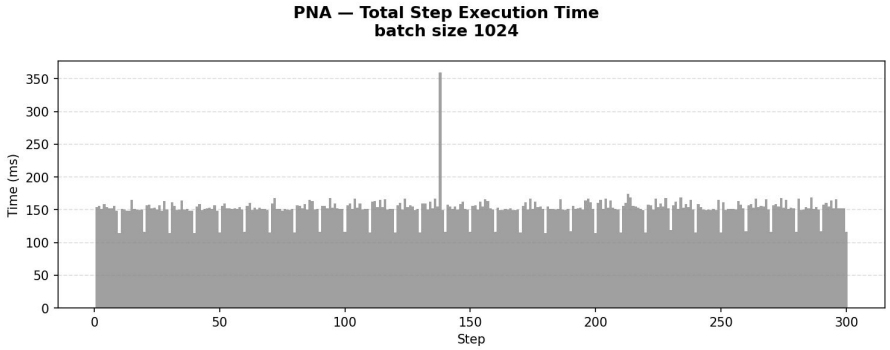
Experiment Setup: Training Configs

Configuration	Batch Sizes	Workers	Purpose
Primary	**4096**	2	Main focus — largest batch that fits in GPU memory. Default batch size in MilaBench.
Secondary	**512**	2	Contrast point — smaller batches expose GC spike patterns and different forward/backward cost balance.
Trend	1024, 2048	2	Fill in the scaling curve between 512 and 4096 for utilization, latency, and energy.
Worker sweep	4096	0, 2, 4	Isolates the effect of DataLoader parallelism on GPU utilization and throughput at fixed batch size.

Raw Measurement



Batchsize: 512



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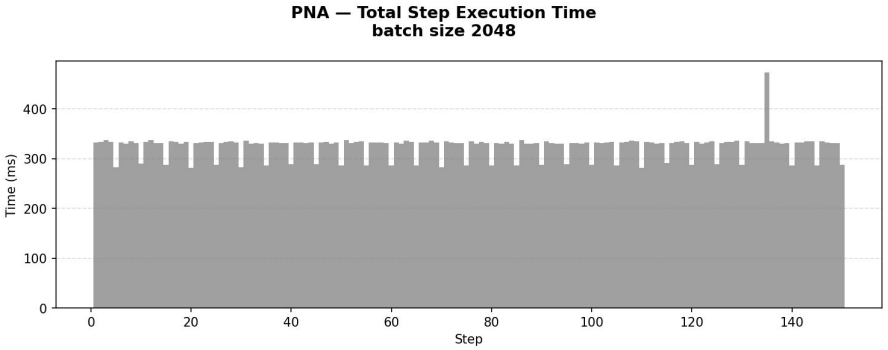
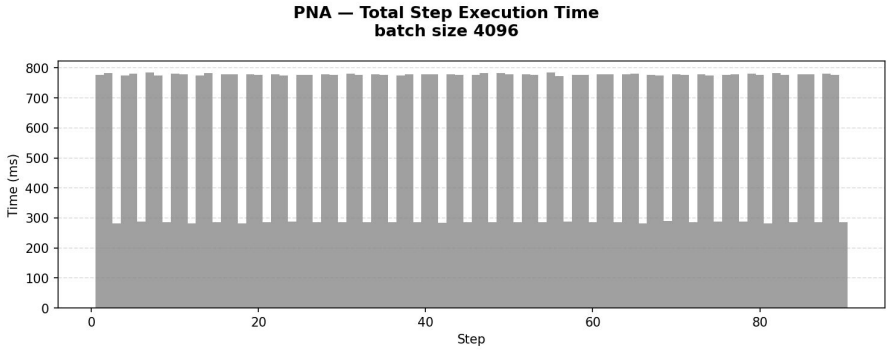
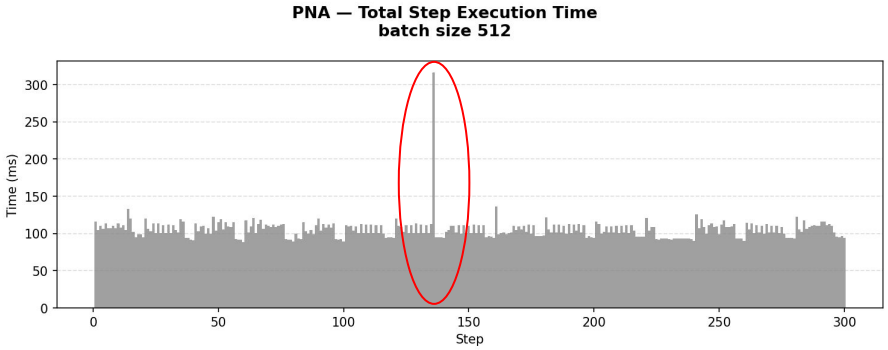


Batchsize: 2048



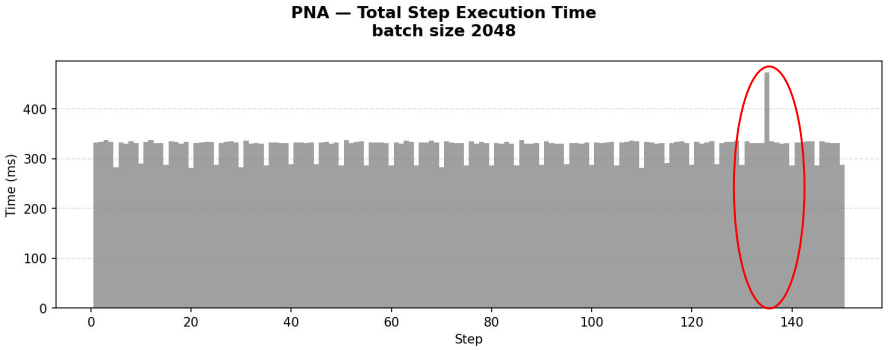
Batchsize: 4096

Raw Measurement



Batchsize: 512

Batchsize: 1024



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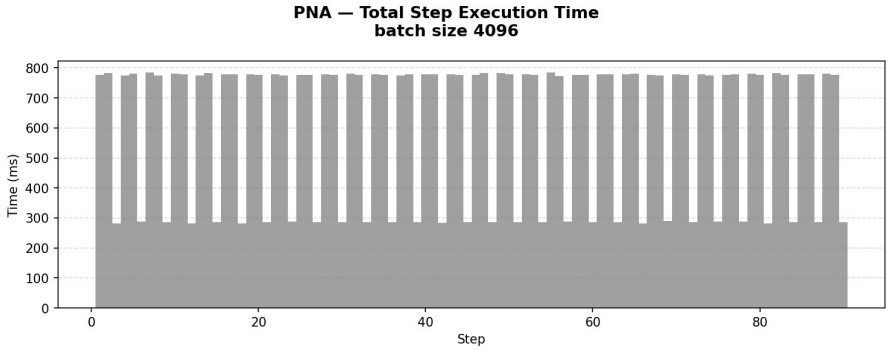
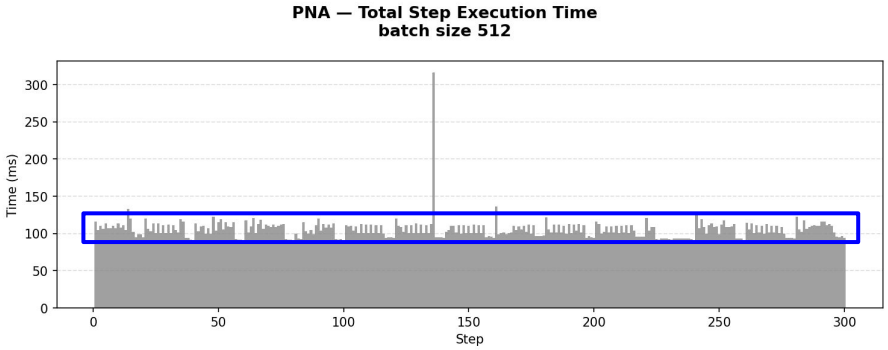


Batchsize: 4096

Raw Measurement



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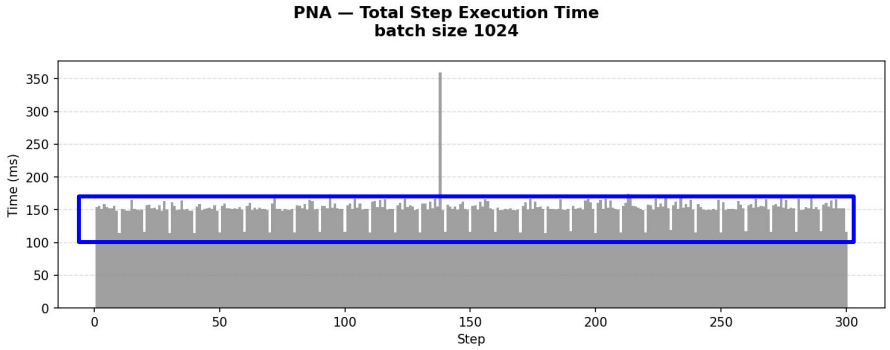

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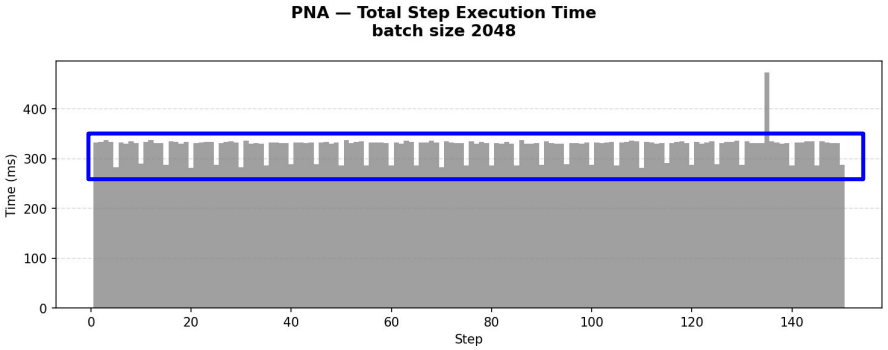

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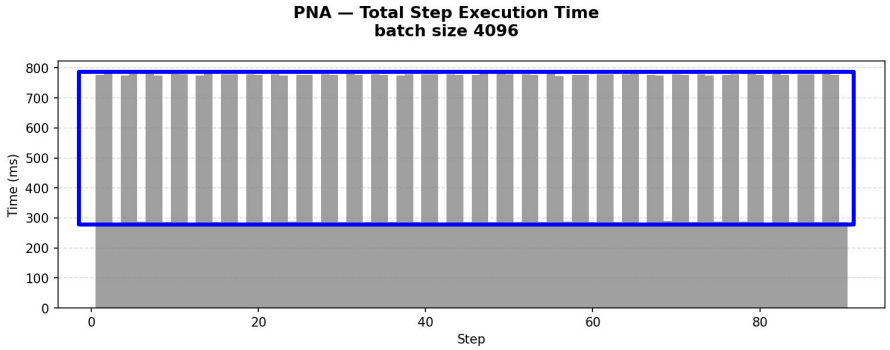
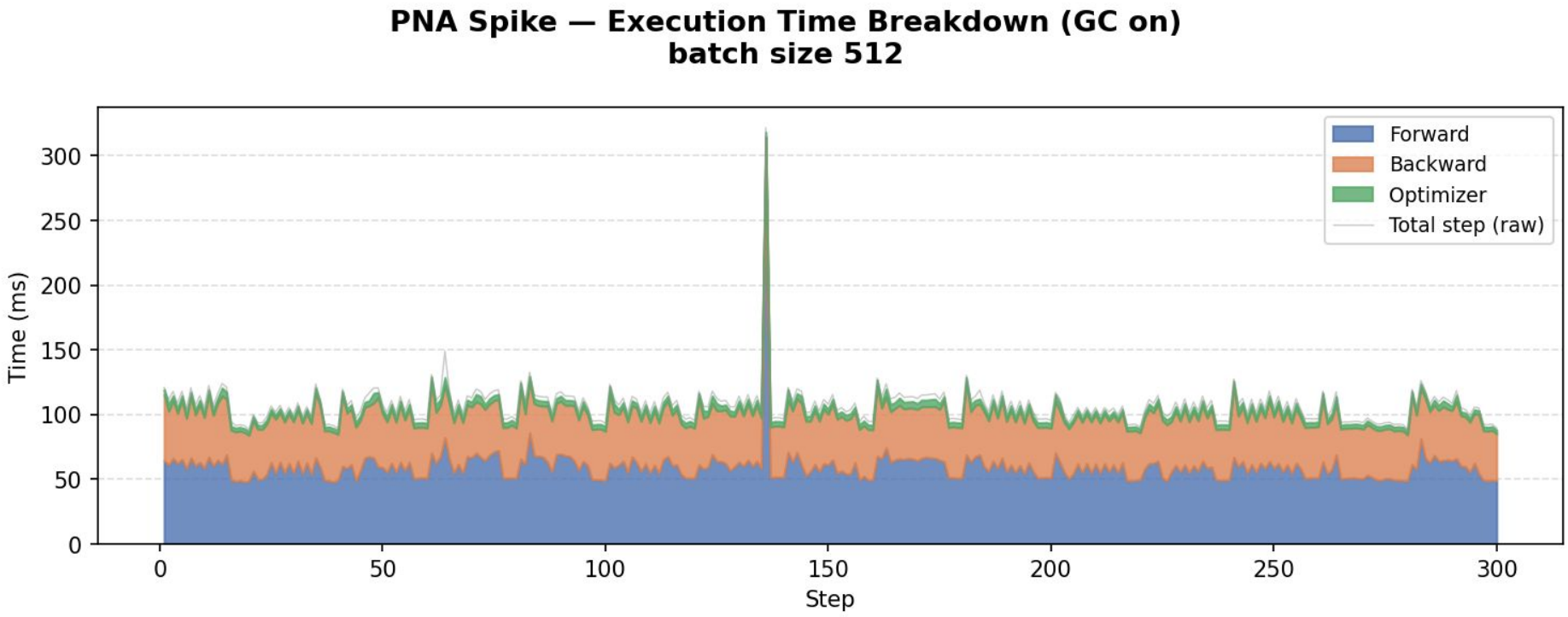


Batchsize: 4096

Spike Analysis: Case 1



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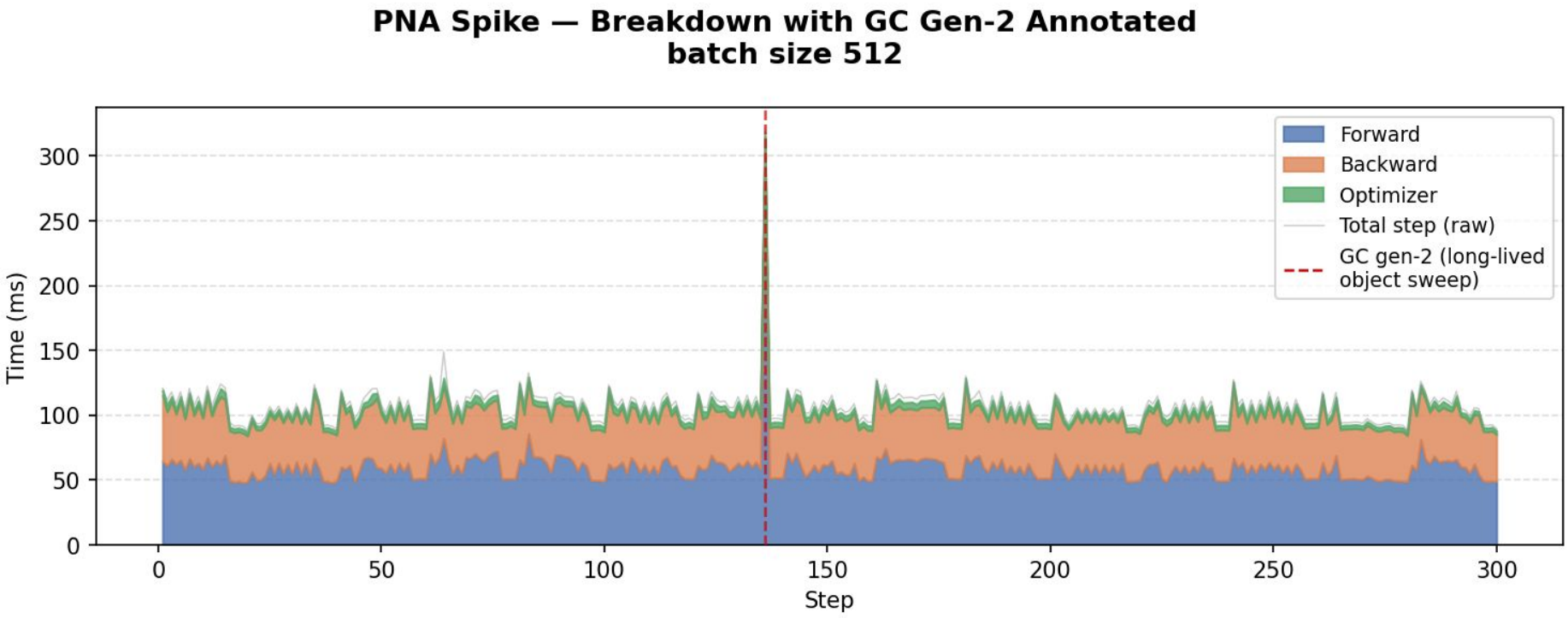

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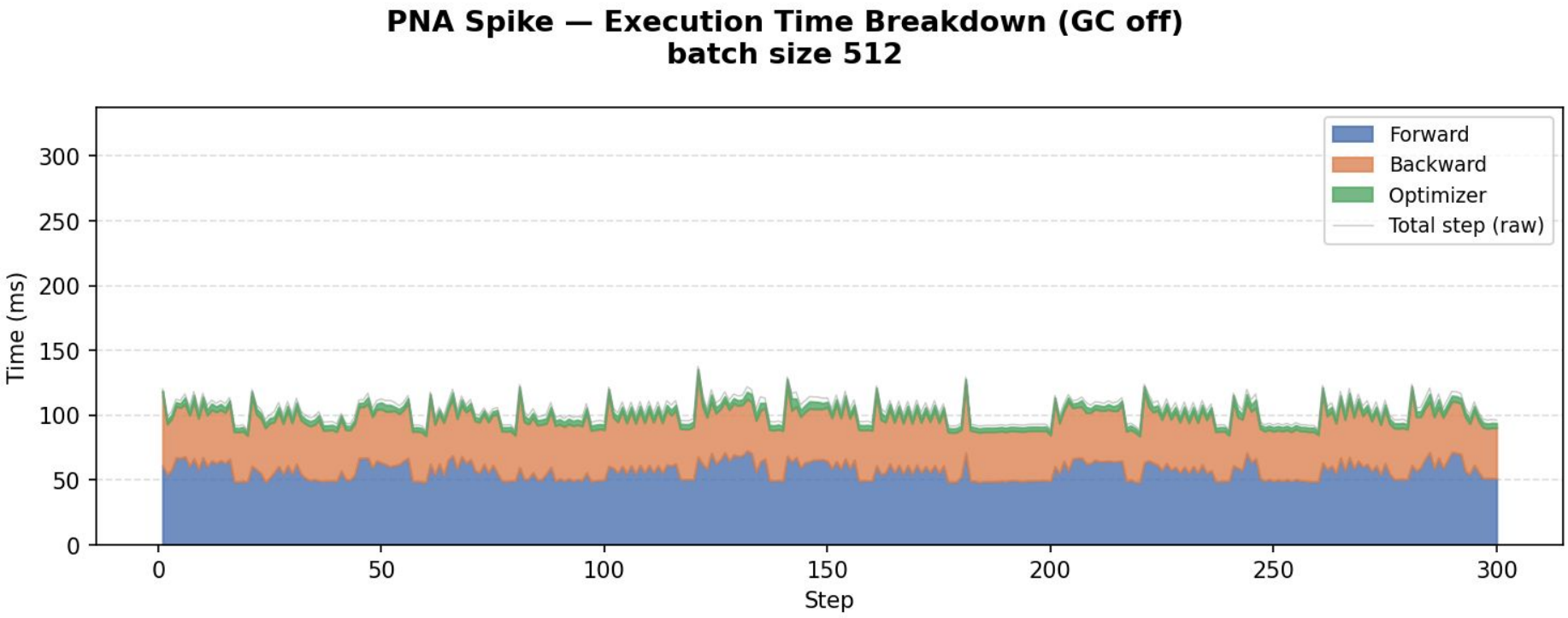
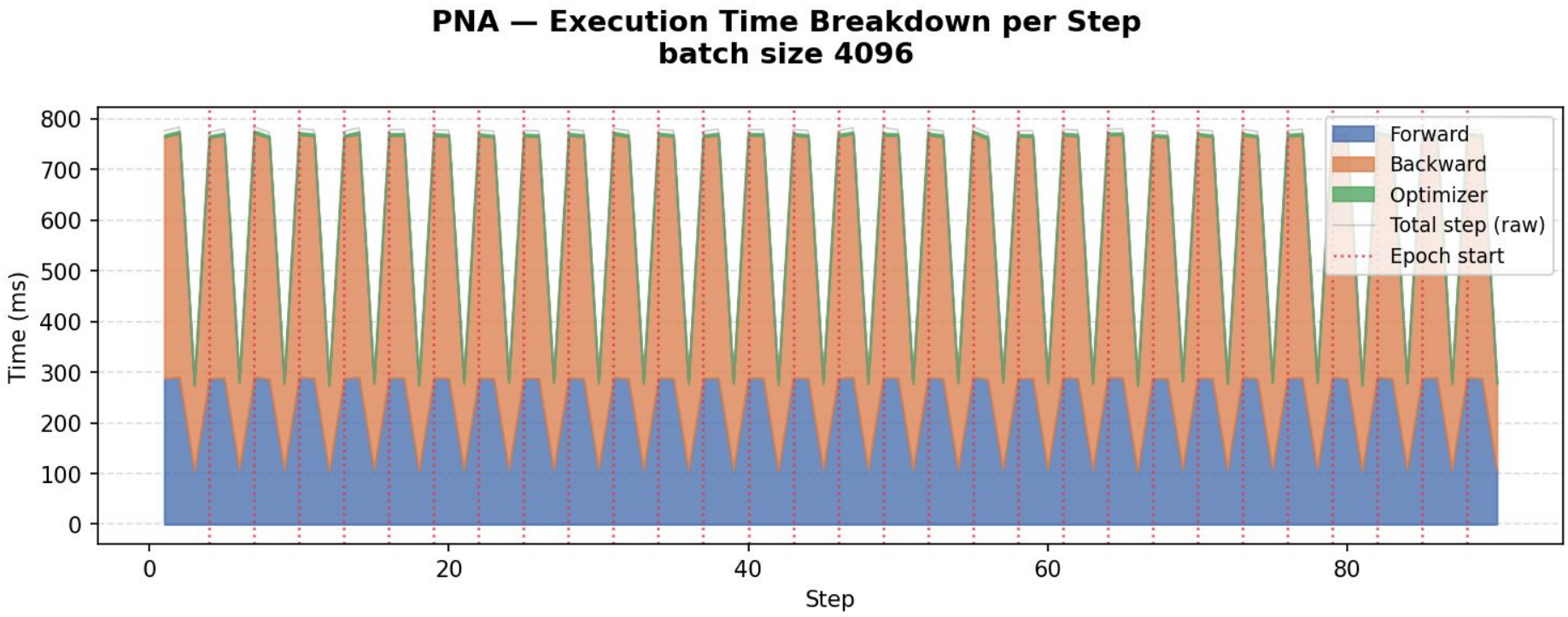
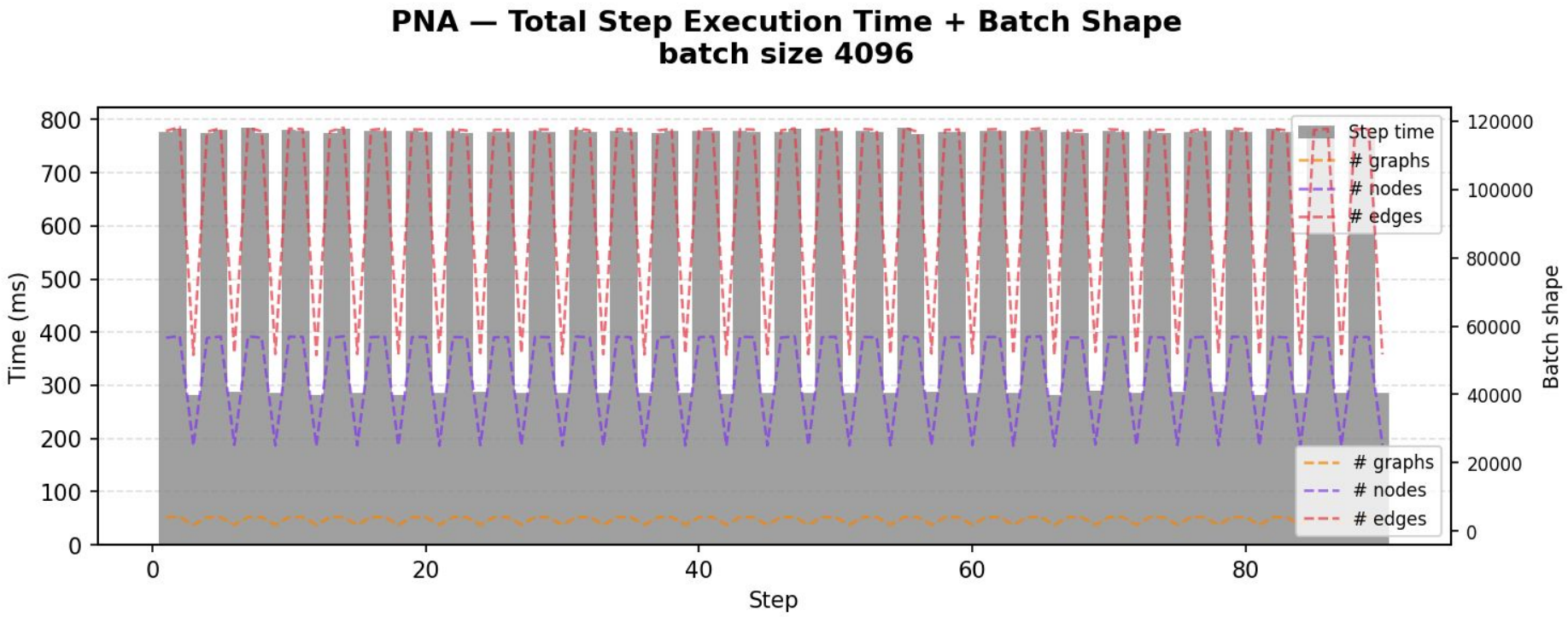


Spike Analysis: Case 2

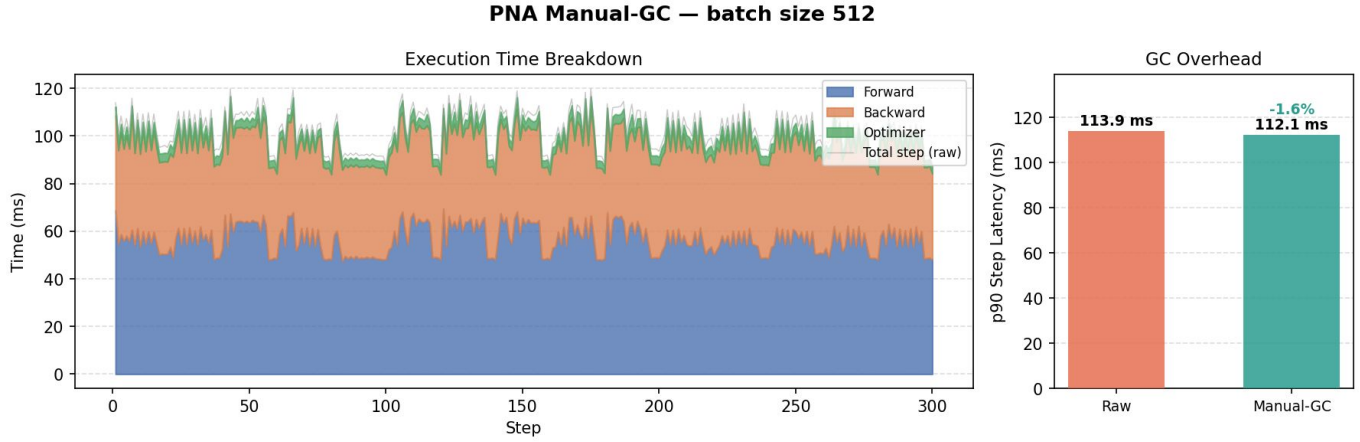


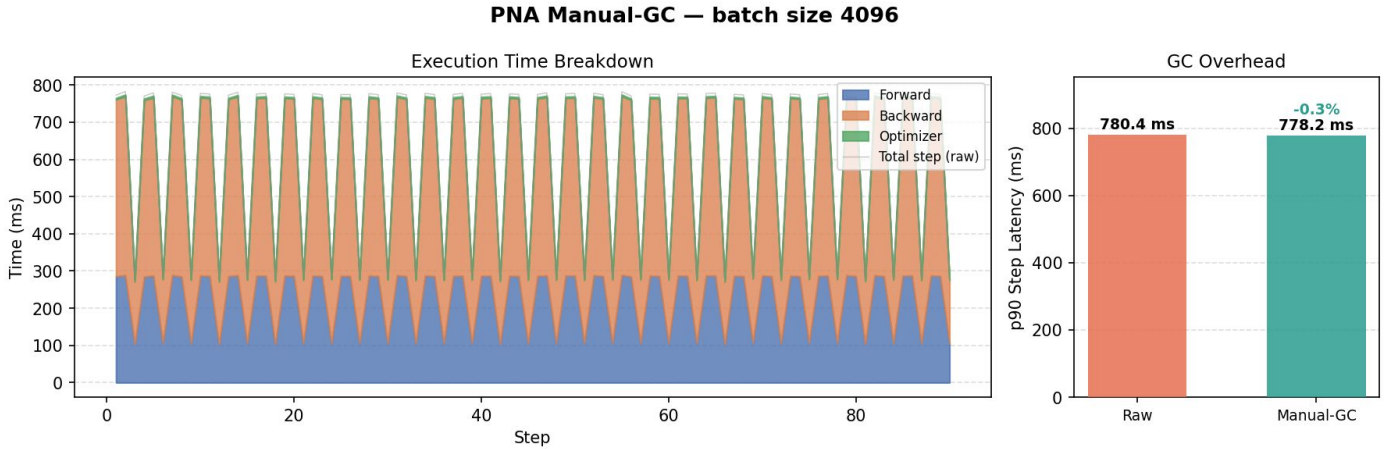
Spike Analysis: Case 2



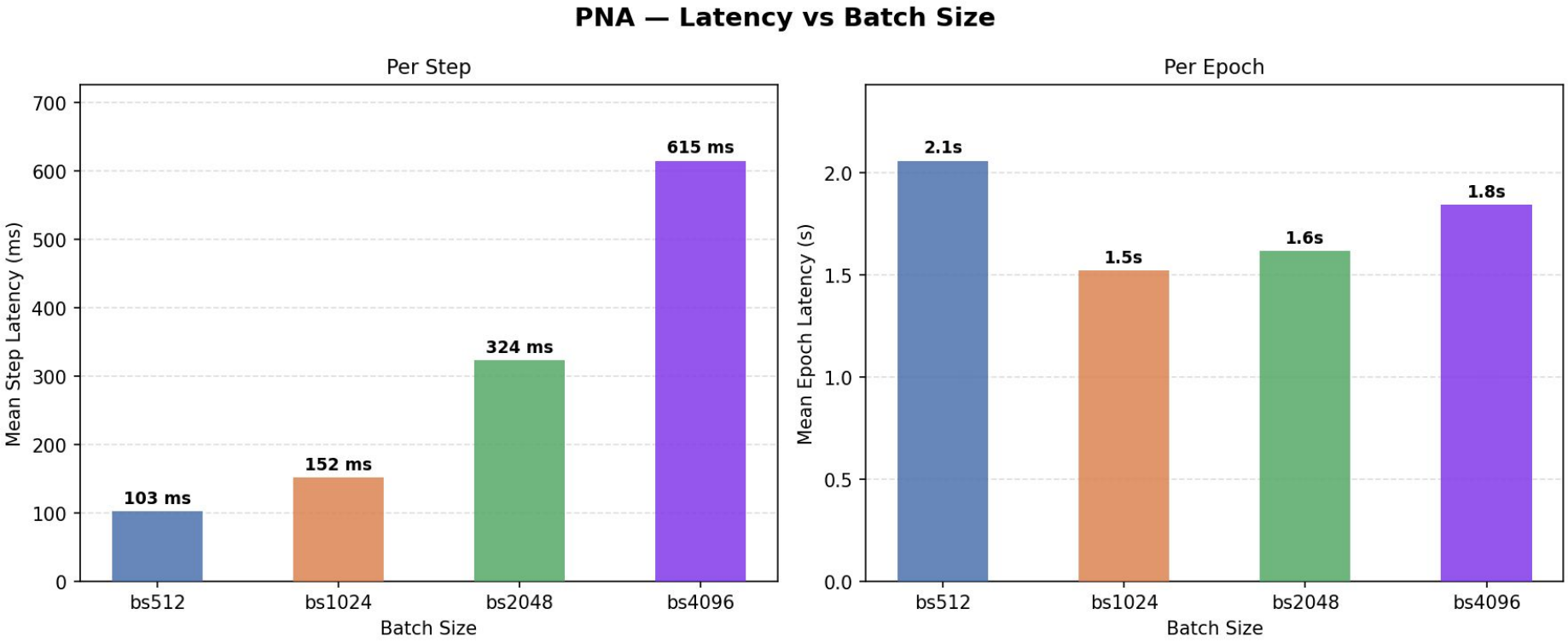
Manual GC Baseline

Establish a clean baseline free of GC noise. Automatic GC disabled during training;
full gen-2 sweep forced between epochs.





Manual GC Baseline: Batch size Comparison



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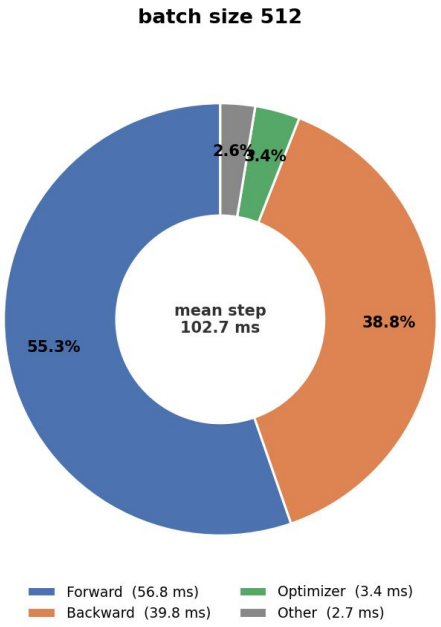
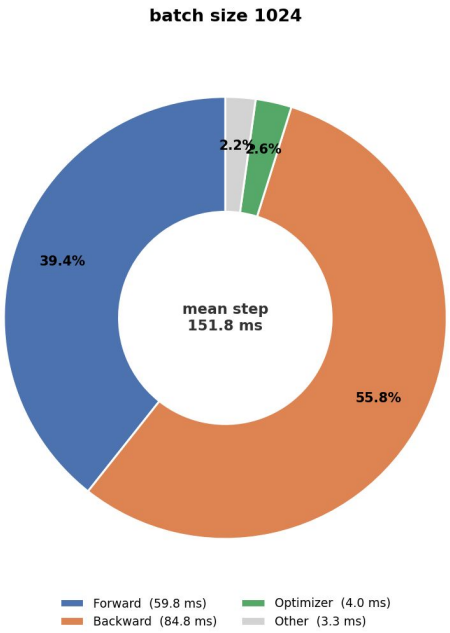
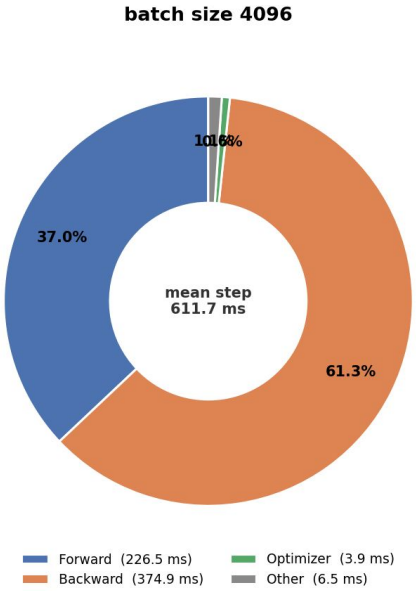
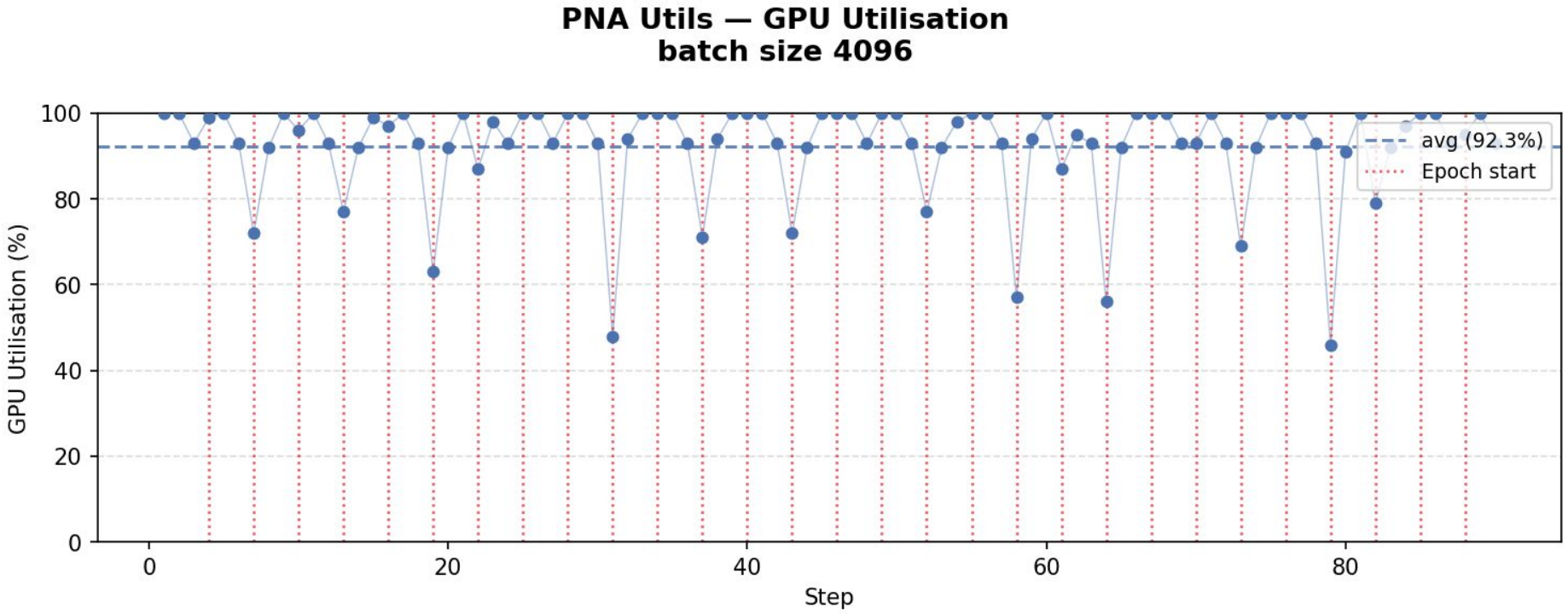


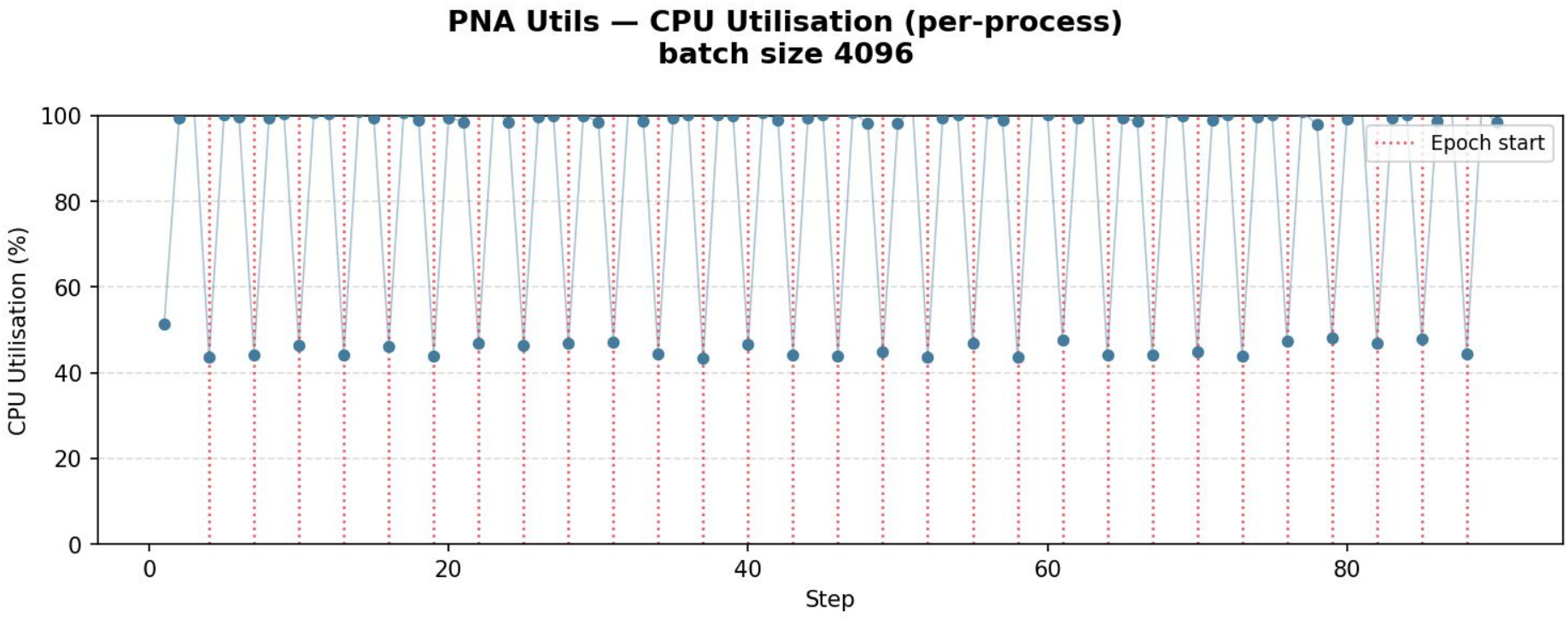




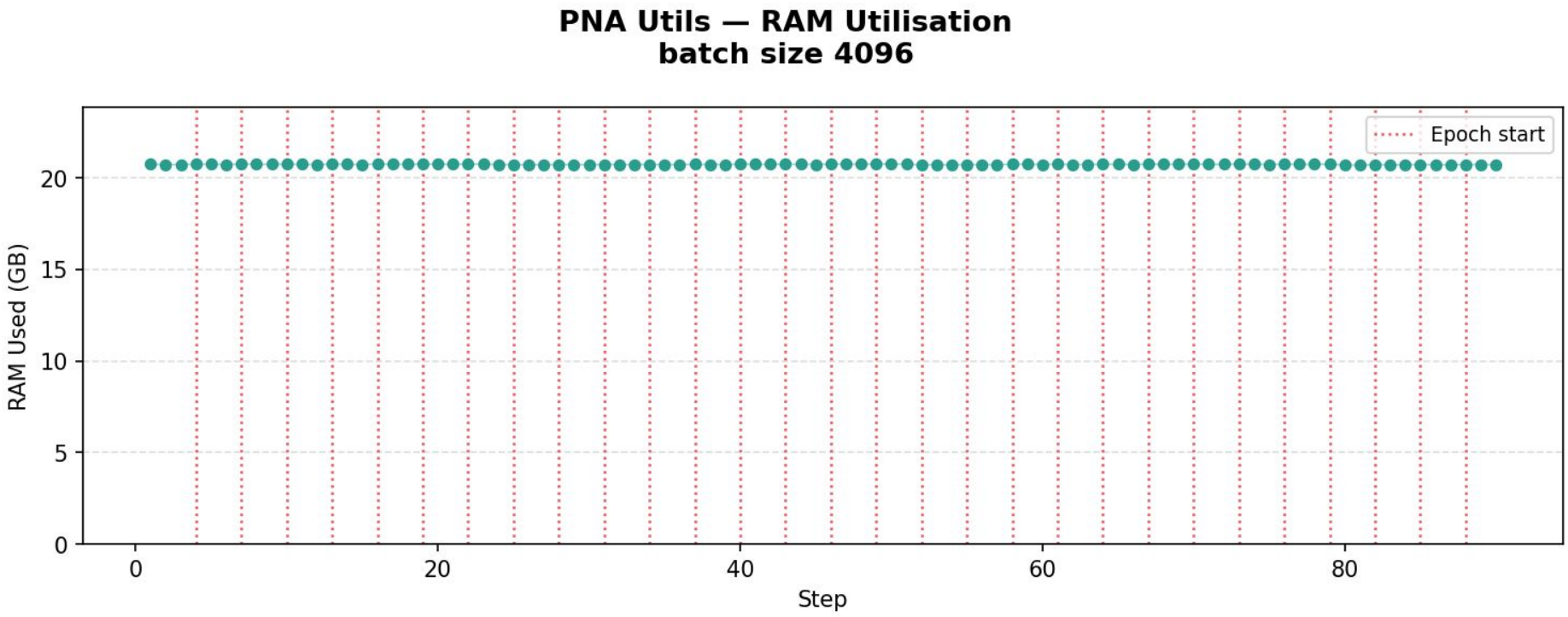
Utilization: Batch Size 4096



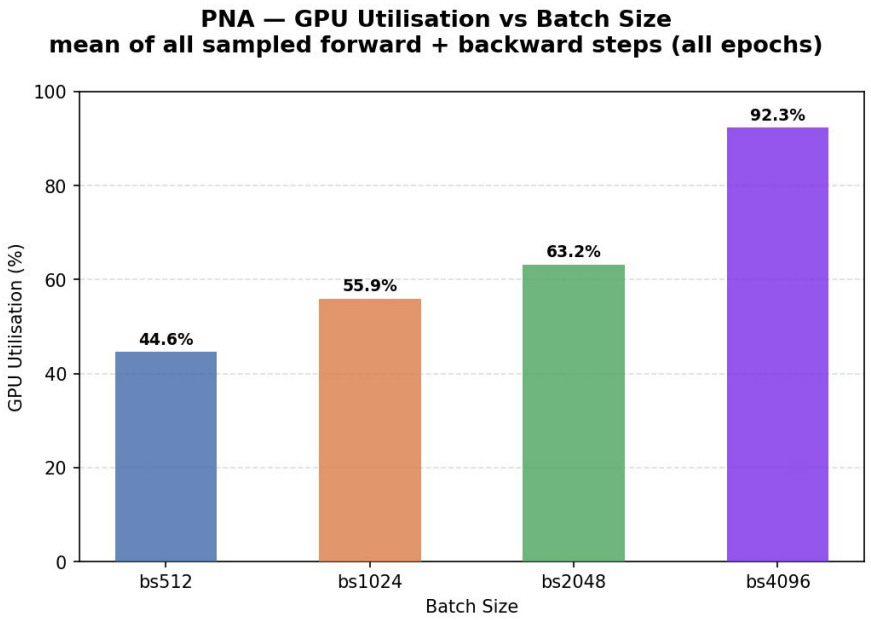
Utilization: Batch Size 4096



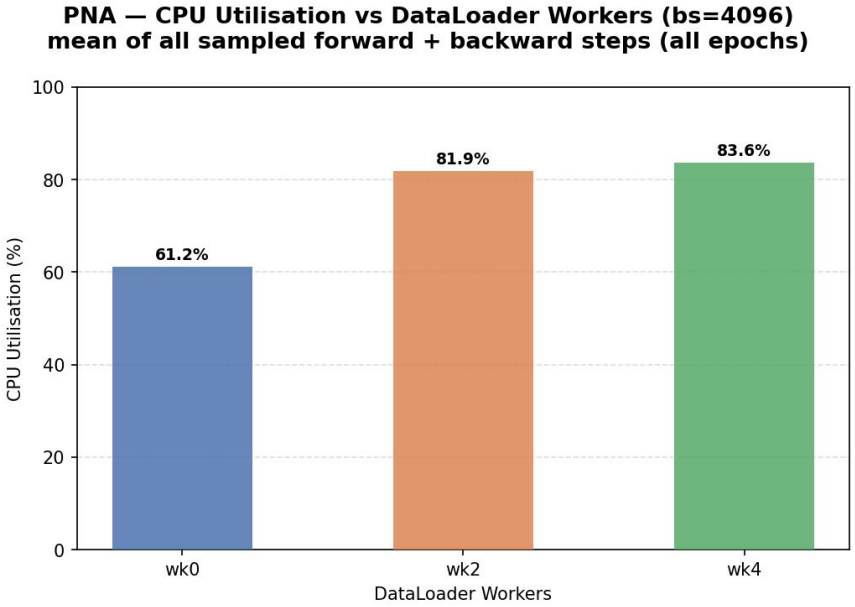
Utilization: Batch Size 4096



Utilization: Varying Batch Sizes

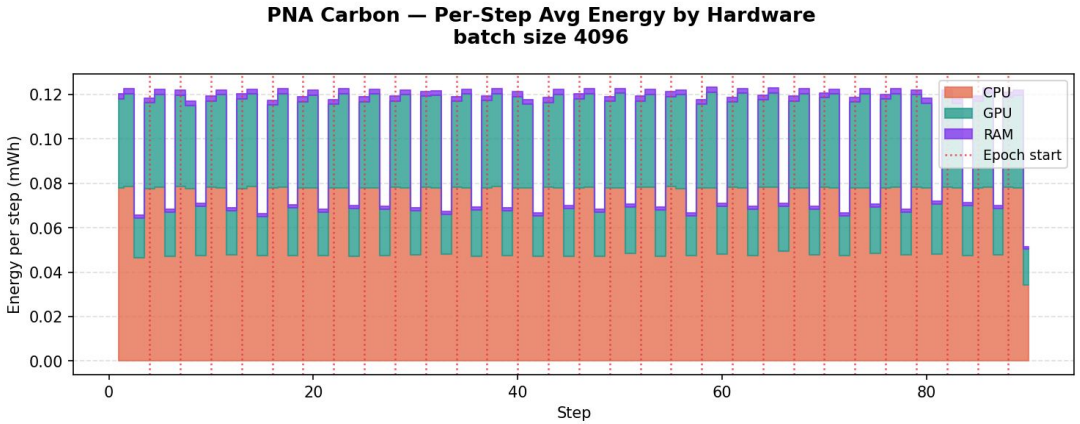


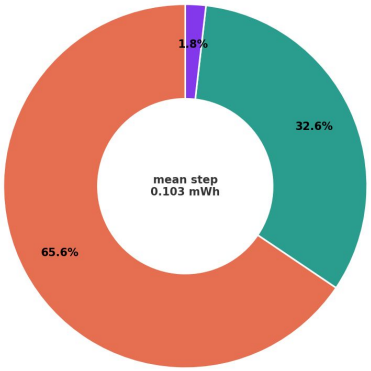
GPU Util

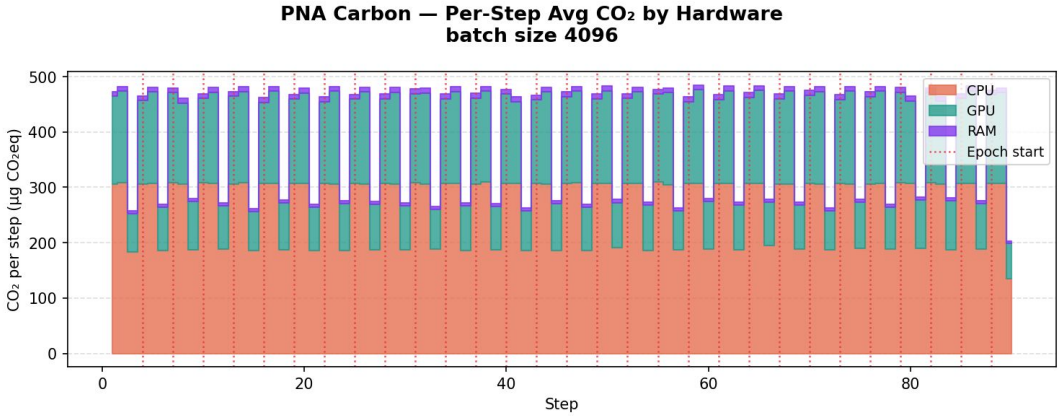


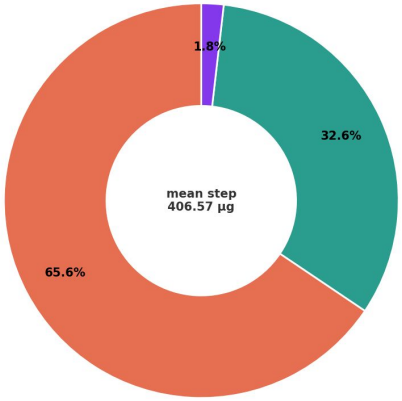
CPU Util

Energy & Carbon: Batch Size 4096

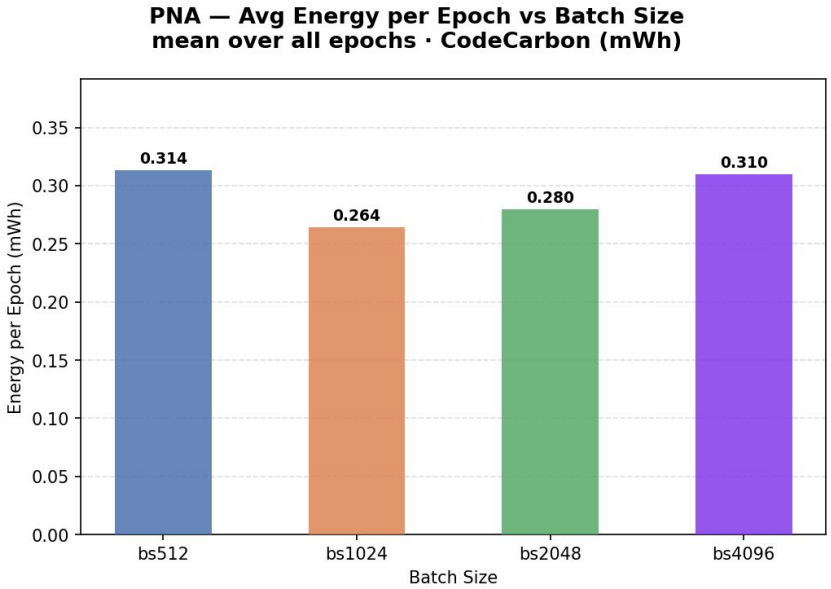




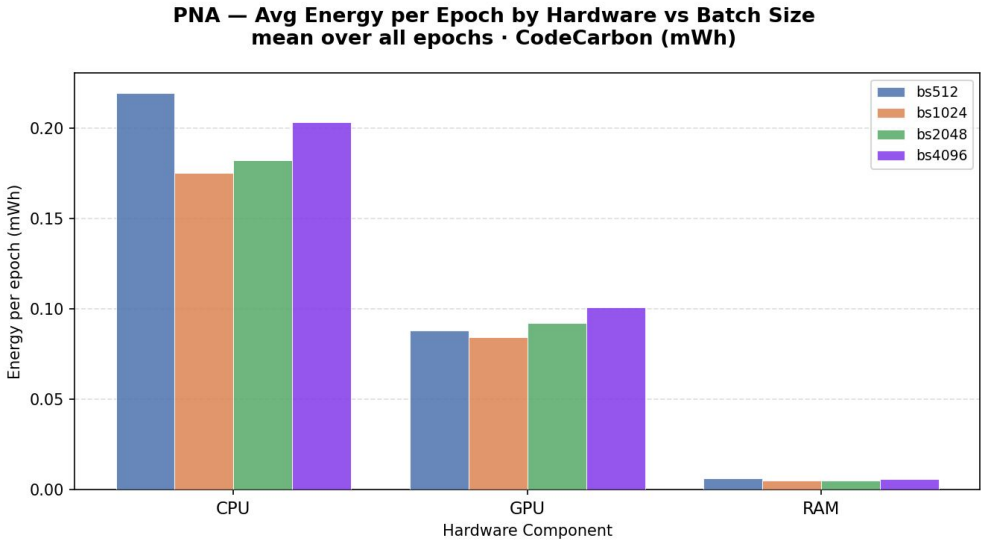




Energy & Carbon: Varying Batch Sizes

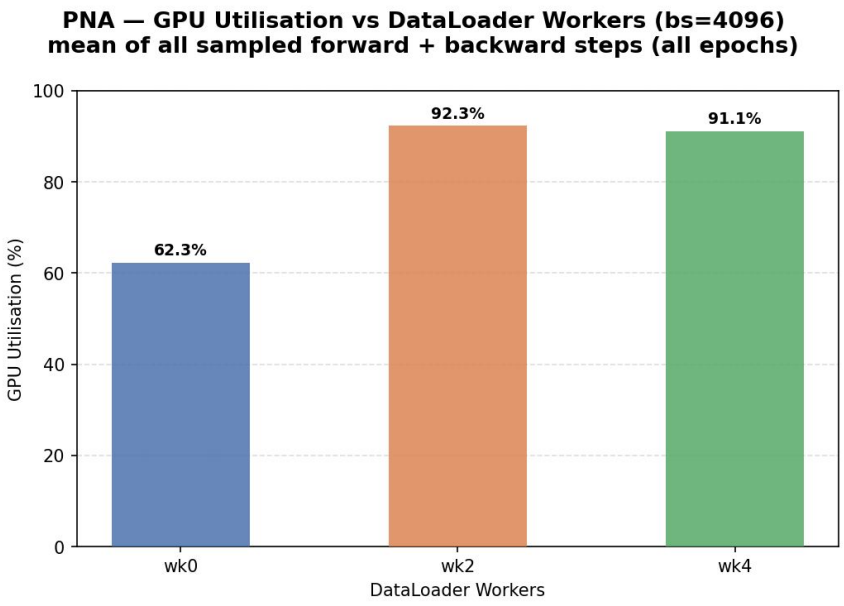


Energy per epoch

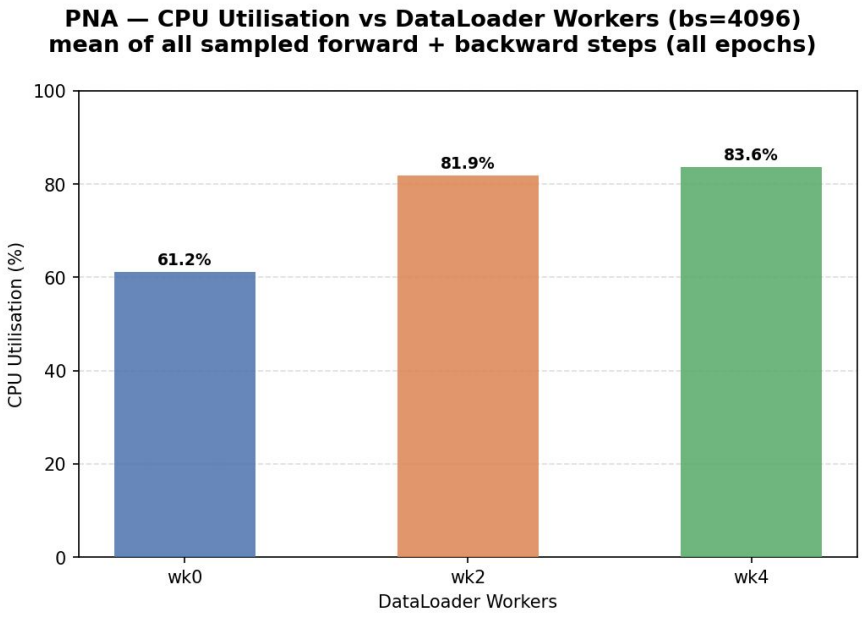


Breakdown by hardware

Varying Workers: Hardware Util

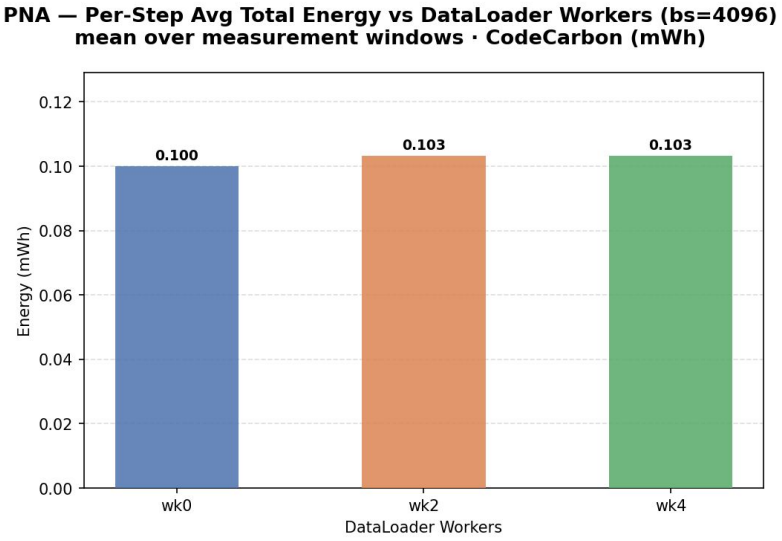


GPU Util Varying #Workers

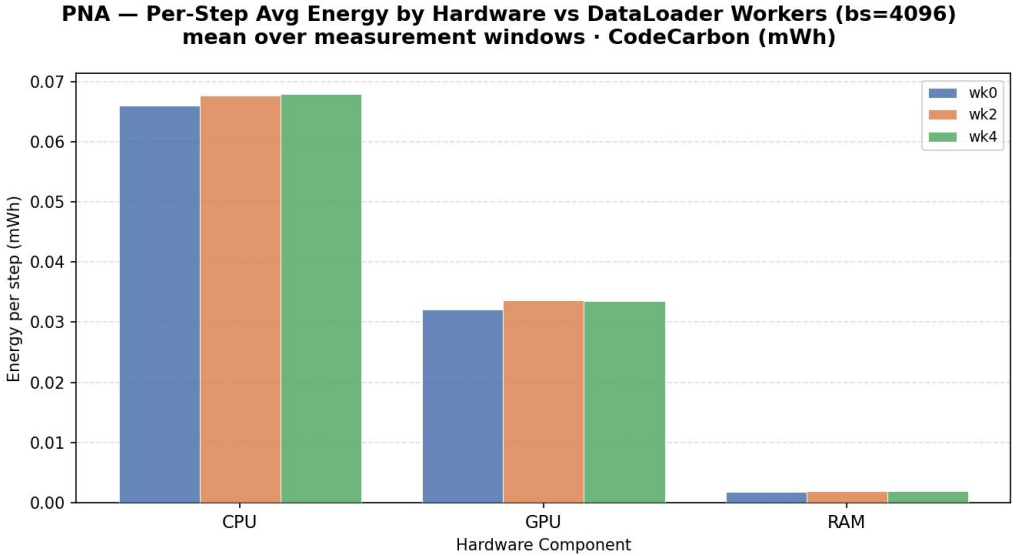


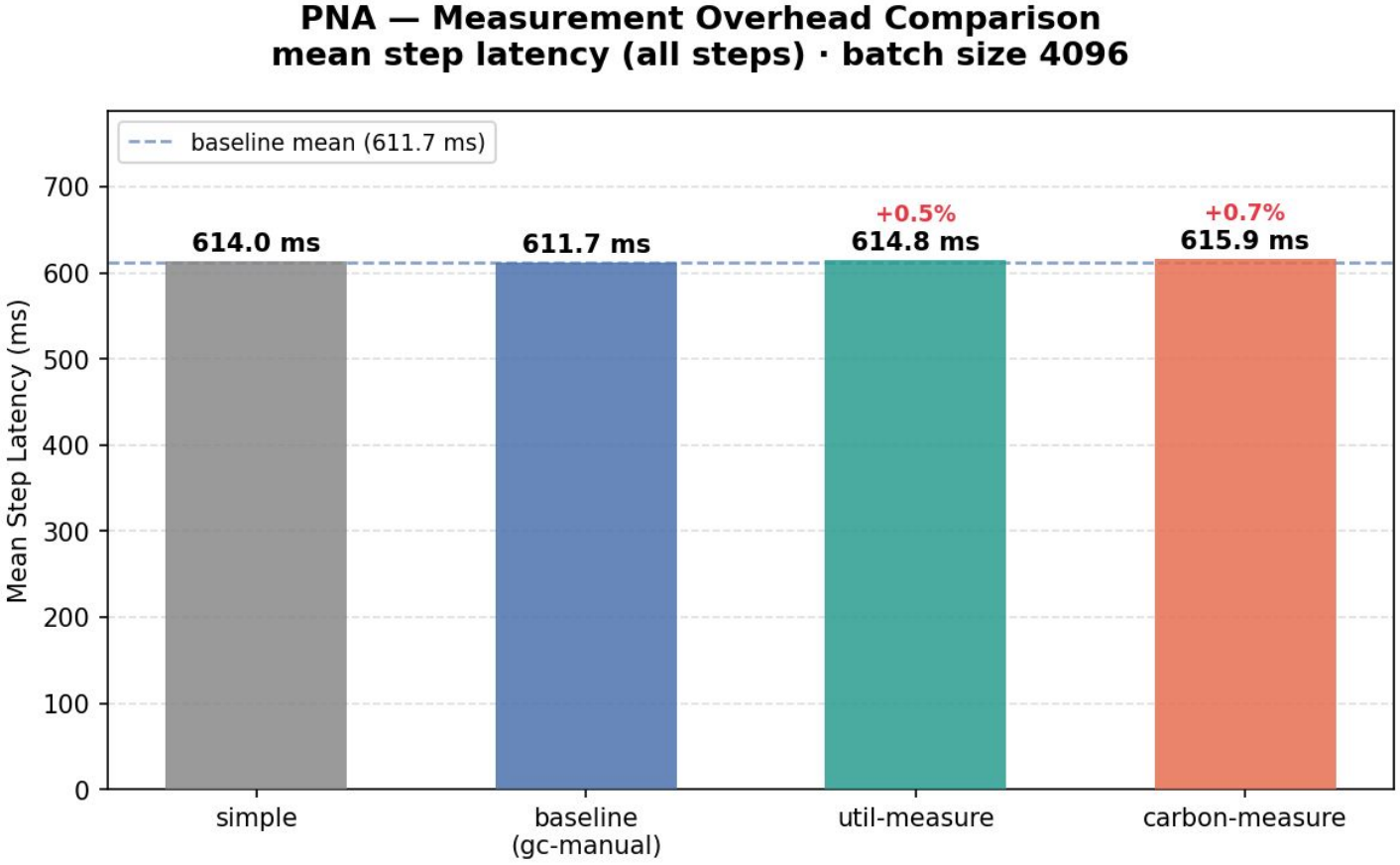
CPU Util Varying #Workers

Varying Workers: Energy



Energy per step



Breakdown by hardware

Measurement Overhead